

# Comparing Policy Sharing Methods in MPE Simple Spread

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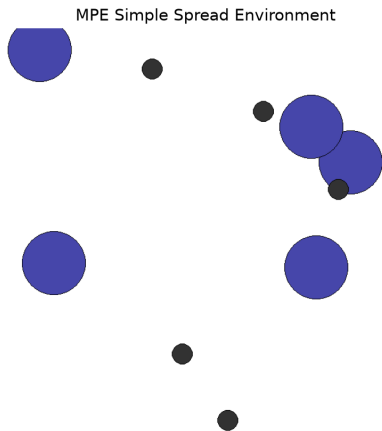
Final Project Presentation

## What I wanted to check

- I used MPE Simple Spread, where several agents have to cover different landmarks.
- I compared three ways of building the policy network:
  - ▶ one shared policy,
  - ▶ one shared policy with agent ID,
  - ▶ separate policies for each agent.
- My main question was whether sharing parameters helps learning when every agent has only partial observations.
- I also checked whether a better return actually meant that the agents completed the task.

**Final comparison:** 5 agents and 5 landmarks. Earlier 2-agent and 3-agent runs were used as smaller test cases.

## Environment used in the final experiment



- 5 agents and 5 landmarks
- Goal: each agent should cover a different landmark
- Actions: stay, left, right, down, up
- Maximum length: 200 steps
- Each agent sees one nearby agent and two nearby landmarks

This setting is still small, but it becomes difficult when agents have to decide who should go to which landmark.

## What changed as I increased the number of agents?

**2 agents**



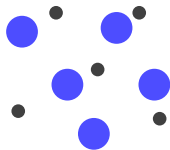
Successful episodes observed

**3 agents**



Successful episodes observed

**5 agents**

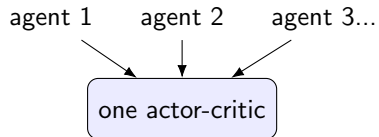


0 success in final evaluation

The 2-agent and 3-agent runs were early checks. The numerical comparison in the rest of this presentation uses the 5-agent setting.

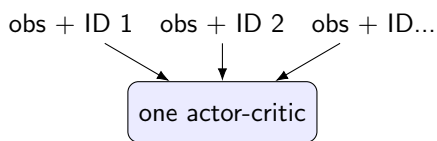
# The three policy structures

## Shared



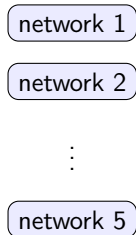
All trajectories update the same network.

## Shared + Agent ID



The same network can react differently to each ID.

## Independent



Each agent learns from its own data.

# How I trained the policies

- PPO with an actor and a critic
- Two hidden layers, 512 units per layer
- Categorical policy for five discrete actions
- Generalized Advantage Estimation
- Local observations were used by both actor and critic
- For shared policies, data from all agents were put into the same PPO batch

$$L^{clip}(\theta) = \mathbb{E} \left[ \min \left( r_t(\theta) \hat{A}_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

I kept the learning algorithm the same and changed only the policy-sharing structure.

# Reward and success condition

## Reward used during training

- one-to-one agent-landmark matching distance
- reward when the matching distance decreased
- extra reward when more landmarks were covered
- bonus when every assigned landmark was covered

## Success used during evaluation

- every agent had to be close to a different landmark
- success radius: 0.10
- an episode was not counted as successful when agents crowded around the same landmark

I measured return and success separately because the distance reward can improve even when the final assignment is incomplete.

# Final experiment settings

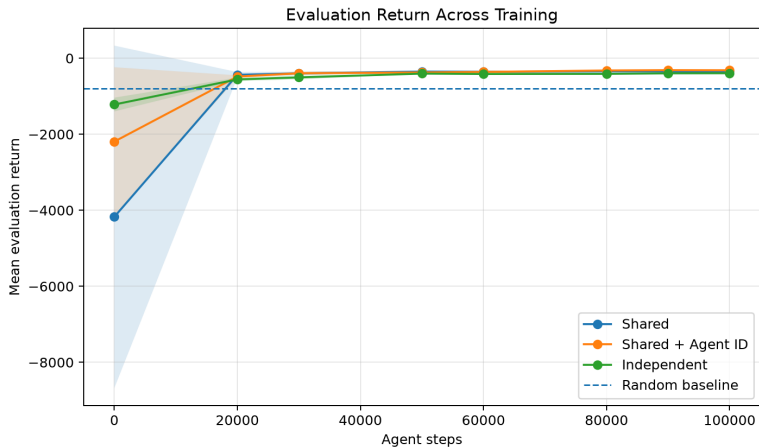
| Setting                | Value              |
|------------------------|--------------------|
| Agents / landmarks     | 5 / 5              |
| Training seeds         | 142, 254           |
| Agent steps per run    | 100,000            |
| Rollout size           | 10,000 agent steps |
| Evaluation interval    | 15,000 agent steps |
| Evaluation episodes    | 20                 |
| Maximum episode length | 200                |
| Discount factor        | 0.995              |
| GAE parameter          | 0.95               |
| PPO clip coefficient   | 0.2                |

## Files saved from the experiment

- learning curves
- final result figures
- CSV summaries
- checkpoints
- evaluation videos

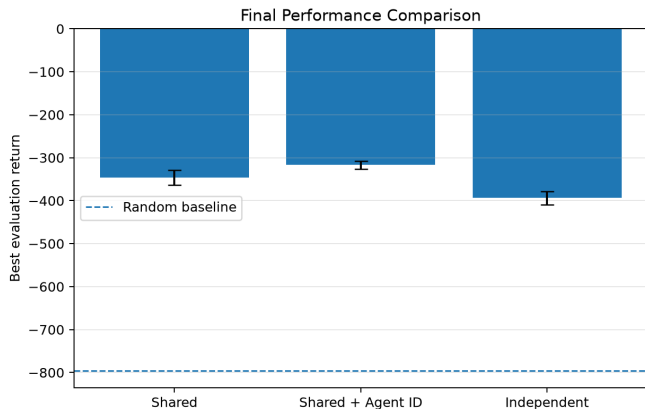
Two seeds are not enough for a strong statistical claim, so I treat this as a project-level comparison.

## Training result: evaluation return



All three methods improved quickly from the random baseline. Most of the change happened in the early part of training.

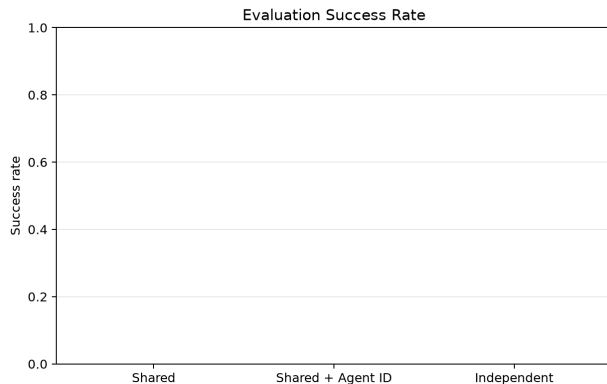
## Best return in the 5-agent setting



| Method      | Return              | Params |
|-------------|---------------------|--------|
| Shared      | $-346.21 \pm 17.33$ | 541k   |
| Shared + ID | $-317.02 \pm 9.14$  | 547k   |
| Independent | $-393.80 \pm 15.26$ | 2.71M  |

Shared + Agent ID had the best average return. The independent version used about five times more parameters, but its return was lower.

## But the 5-agent task was not completed



- Success rate was 0 for all three methods in the final evaluation.
- Agents usually moved toward landmarks, so the return improved.
- The remaining problem was the one-to-one assignment.
- Two agents sometimes approached the same area while another landmark remained empty.

This was different from the smaller 2-agent and 3-agent tests, where I had observed successful episodes.

## What I saw in the evaluation videos



The movement was clearly better than a random policy, but crowding and unstable target selection still appeared in every method.

## Why did scaling from 3 to 5 agents become harder?

- With more agents, there are more possible assignments between agents and landmarks.
- Each agent still sees only a small part of the environment.
- The local critic does not know whether another agent is already moving toward the same landmark.
- Agent ID tells the network which agent is acting, but it does not tell the agent which landmark should be its target.
- The shaped reward gives useful movement signals, but the final coordination decision is still difficult.

My result suggests that the main failure was not basic movement. It was maintaining different roles as the team became larger.

## What I learned from the comparison

- ① **Sharing parameters helped in this experiment.** Both shared methods used much fewer parameters and obtained better return than the independent method.
- ② **Agent ID gave a small improvement.** It produced the best return, but it did not solve target assignment.
- ③ **Return was not enough.** The agents could reduce distance without completing the full task.
- ④ **The number of agents mattered.** Successful behavior seen with 2 and 3 agents did not continue in the 5-agent experiment.

## Limits of this experiment

- The final numerical comparison used only two seeds.
- I tested one environment and one main reward design.
- Hyperparameters were not tuned separately for each method.
- Both the actor and critic used local observations.
- The 2-agent and 3-agent runs were preliminary tests, so they were not performed under exactly the same final evaluation protocol.
- Because the 5-agent success rate was 0, the results compare learning behavior rather than a solved task.

## Changes I would try next

- assign a target landmark to each agent at the beginning of an episode
- give each agent a progress reward for its assigned target
- use a centralized critic during training while keeping decentralized actors for execution
- add a recurrent network for partial observations
- run more seeds and compare 2, 3, 4, and 5 agents with the same protocol
- test selective sharing instead of forcing all agents to share or all agents to be independent

# Conclusion

- In the 5-agent experiment, Shared + Agent ID produced the best return:  $-317.02$ .
- The shared methods needed far fewer parameters than independent policies.
- However, none of the 5-agent methods completed the task during final evaluation.
- Earlier 2-agent and 3-agent tests did produce successful episodes.
- The main problem after scaling up was stable role assignment, not simply moving toward landmarks.

Thank you.

# References

- R. Lowe et al., “Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments,” NeurIPS, 2017.
- J. Schulman et al., “Proximal Policy Optimization Algorithms,” arXiv, 2017.
- J. K. Terry et al., “Parameter Sharing is Surprisingly Useful for Multi-Agent Deep Reinforcement Learning,” 2020.
- J. K. Terry et al., “PettingZoo: Gym for Multi-Agent Reinforcement Learning,” NeurIPS, 2021.